Midterm Project

1 Overview

During the first half of this course, we explored methods for preparing, visualizing, and analyzing data to extract meaningful insights. This midterm project asks you to apply these skills in an open-ended context by analyzing a real dataset.

Unlike structured labs, this project requires you to determine appropriate preprocessing steps, visualizations, and statistical tests. Your goal is to construct a clear and evidence-based argument addressing the research questions below.

2 Background

Conservation ecologists monitor animal populations to evaluate ecosystem health and the effectiveness of conservation strategies. Changes in population trends can indicate environmental shifts, emerging threats, or successful management practices.

Several methods exist for estimating wildlife populations. Traditional approaches include mark–recapture studies and camera-trap surveys. In addition to these techniques, researchers sometimes analyze biological remnants such as scat or carcasses to infer species presence.

In a 2015 study, Dr. Rachel Reid investigated whether morphological (e.g., size and shape), biogeochemical (e.g., chemical composition), and contextual (e.g., geographic location) traits could be used to distinguish between bobcat, coyote, and fox scat samples. Morphological traits are typically the most practical for field identification, while biogeochemical traits require laboratory analysis.

You will analyze Dr. Reid's dataset to evaluate whether specific traits can reliably differentiate these species.

Research Questions

1. Which morphological or biogeochemical traits distinguish between species?
2. What biological or ecological factors might explain these differences?

3 Project Tasks

Your written report should follow the structure of a short research paper.

The report should contain the following sections:

- Introduction
- Methods
- Results
- Discussion

The tasks described below correspond to the content expected in each section of the report.

Part I: Introduction and Background

1. Read the provided article by Dr. Reid.
2. Research the three species (bobcat, coyote, and fox), focusing on their diet, habitat, and physical characteristics.
3. Find one scholarly article discussing methods used to estimate wildlife populations in conservation ecology.
4. Write an introduction that addresses:
    - The ecological significance of monitoring these species.
    - Why morphological and biogeochemical traits may serve as useful identifiers.
    - Key biological differences between the species that may influence scat characteristics.
    - Your hypotheses regarding the research questions.

Part II: Dataset Documentation (Data Card)

Before performing your analysis, you must create a **dataset card** using the format discussed in class. This document is separate from the report and is intended to help you understand and document the dataset prior to analysis.

You will submit the dataset card as a **separate document**. It does not count toward the page limit of the report, but the information you develop here should directly inform the methods and analysis described in your report.

Follow the provided template and complete all relevant sections.

At minimum, your dataset card should include:

- A brief summary of the dataset and its origin.

- A table describing each feature, including its data type and a short description. You should also categorize each feature as *morphological*, *biogeochemical*, *contextual*, or *not a trait*.
- Summary statistics for numerical variables and basic summaries for categorical variables.
- A small set of exploratory visualizations that help describe the structure of the dataset.
- Notes describing any notable preprocessing steps, missing data, or potential limitations of the dataset.

Part III: Methods

Describe the analytical steps you used.

1. **Data preprocessing**
   - Describe how the dataset was cleaned.
   - Explain how missing values were handled.
   - Describe any transformations or type conversions performed.
   - Explain how outliers were treated (removed, transformed, or retained).
2. **Statistical and visualization methods**
   - Describe the statistical tests used to evaluate differences between species.
   - Explain why those tests were appropriate.
   - Describe the visualizations used and their purpose.

Part IV: Analysis

1. Load and clean the dataset. Ensure that all columns are assigned appropriate data types.
2. Use visualizations and statistical tests to evaluate relationships between species and the **morphological and biogeochemical traits**. Do **not** analyze contextual variables.
3. Identify which traits distinguish at least one species from the others. Support your findings with appropriate visualizations and statistical evidence.
4. Include only figures that contribute to your argument.

Part V: Discussion and Interpretation

1. Create a summary table listing traits that distinguish species. Include descriptive statistics (e.g., mean, median, or distribution summaries).
2. Using your biological research, explain why these traits may differ between species.
3. Explain why morphological or biogeochemical traits may be more useful than contextual traits for species identification.

4 Submission and Grading

- **This is an individual assignment.** You may not collaborate or discuss the midterm project with others.
- Submit the following three items:
  - **Report (PDF)** – A short research-style report addressing the project tasks. This report should be approximately **2 pages in length**, including figures, and excluding references.
  - **Dataset Card (PDF)** – Your completed dataset card following the template discussed in class. This document should be submitted **separately** and does **not** count toward the 2-page report limit.
  - **Jupyter Notebook (.ipynb)** – The notebook used to perform your analysis.
- The report should present a clear argument addressing the research questions and should include appropriately labeled figures and tables where necessary.
- All figures must be referenced in the text of the report (e.g., "Figure 1 shows . . . "). Figures that are included but not referenced in the text will be ignored during grading.
- Evaluation will consider:
  - Appropriateness of analytical methods
  - Correctness of statistical reasoning
  - Quality of interpretation and discussion
  - Clarity of visualizations and presentation
  - Completeness and quality of the dataset card
- **Late policy**: 5% deducted per day for up to three days. Submissions are not accepted after that.